

Physical Chemistry–III (CHB-504)

Unit 5: Electrolyte Solutions

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This is hard

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Why Electrolyte Solutions Are Non-Ideal

Electrolyte solutions deviate strongly from ideal behaviour even at very low concentrations because ions carry **electric charges**. Long-range electrostatic (Coulombic) interactions ($\propto 1/r$) between ions persist at much greater distances than the short-range Van der Waals interactions between neutral molecules. These ion–ion interactions lower the Gibbs energy of the solution below the ideal value, causing the mean activity coefficient $\gamma_{\pm} < 1$.

Consequences of ion–ion interaction:

1. Each ion is surrounded by an **ionic atmosphere** of opposite charge.
2. The chemical potential (and activity) of every ion is reduced relative to its ideal value.
3. Transport properties (conductance, diffusion) are retarded by the ionic atmosphere through the *relaxation effect* and *electrophoretic effect*.
4. Colligative properties of electrolyte solutions are larger than predicted for non-electrolytes (van't Hoff factor $i > 1$).

Activity and Mean Activity Coefficient

Activity and Chemical Potential

For a real solution, the chemical potential of ion i is written:

$$\mu_i = \mu_i^\circ + RT \ln a_i$$

where a_i is the **activity**. For ideal solution: $a_i = c_i/c^\circ$ (or m_i/m°). For a real solution:

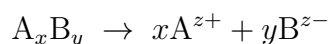
$$a_i = \gamma_i \frac{m_i}{m^\circ}$$

where γ_i is the **activity coefficient** of ion i ($\gamma_i \rightarrow 1$ as $m \rightarrow 0$).

Mean Ionic Activity and Mean Activity Coefficient

Individual ionic activity coefficients γ_+ and γ_- cannot be measured separately (single-ion activities are not directly observable). Instead, the **mean ionic activity coefficient** $\gamma_{\pm}$ is defined.

For an electrolyte A_xB_y dissociating as:



The mean ionic molality and mean activity coefficient are:

$$m_{\pm} = (m_+^x m_-^y)^{1/\nu}, \quad \nu = x + y$$

$$\gamma_{\pm} = (\gamma_+^x \gamma_-^y)^{1/\nu}$$

$$a_{\pm} = \gamma_{\pm} m_{\pm} / m^\circ$$

The mean activity of the electrolyte:

$$a_{\pm}^{\nu} = a_{+}^x a_{-}^y$$

Why geometric mean? The chemical potential involves $\ln a$, so the geometric mean (product of powers) correctly combines contributions from both ions into a single measurable quantity.

Ionic Strength

Definition

The **ionic strength** I of a solution is defined by:

$$I = \frac{1}{2} \sum_i c_i z_i^2$$

where the sum runs over *all* ionic species, c_i is the molar concentration (mol L^{-1}) and z_i is the charge number of ion i .

The ionic strength weighs each ion by the *square* of its charge, reflecting the fact that the electrostatic energy scales as z^2 .

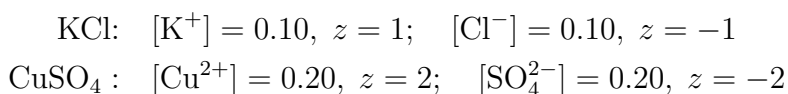
Physical Significance

I is a measure of the total *electrostatic environment* in a solution. The Debye–Hückel theory shows that $\ln \gamma_{\pm} \propto -\sqrt{I}$, so a single parameter I captures the combined effect of all ionic species on the activity coefficient.

Worked Examples

Example 1: KCl + CuSO₄

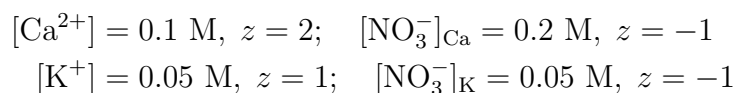
Solution is 0.10 mol kg^{-1} in KCl and 0.20 mol kg^{-1} in CuSO₄ (assuming complete dissociation):



$$I = \frac{1}{2} [0.10(1)^2 + 0.10(1)^2 + 0.20(4) + 0.20(4)] = \frac{1}{2} [0.10 + 0.10 + 0.80 + 0.80] = \frac{1}{2} (1.80) = \boxed{0.90 \text{ mol kg}^{-1}}$$

Example 2: Ca(NO₃)₂ + KNO₃ (Mixed Solution)

50 mL of 0.2 M Ca(NO₃)₂ + 50 mL of 0.1 M KNO₃ (total volume 100 mL; concentrations halved):

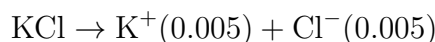


Total $[\text{NO}_3^{-}] = 0.2 + 0.05 = 0.25 \text{ M}$:

$$I = \frac{1}{2} [0.1(4) + 0.25(1) + 0.05(1)] = \frac{1}{2} [0.4 + 0.25 + 0.05] = \frac{1}{2} (0.70) = \boxed{0.35 \text{ mol L}^{-1}}$$

Example 3: $\text{BaCl}_2 + \text{KCl}$

0.008 molal BaCl_2 + 0.005 molal KCl :



$$I = \frac{1}{2}[0.008(4)+0.021(1)+0.005(1)] = \frac{1}{2}[0.032+0.021+0.005] = \frac{1}{2}(0.058) = \boxed{0.029 \text{ mol kg}^{-1}}$$

Debye–Hückel Theory

General Features of the Model

1. The electrolyte is **completely dissociated** into ions.
2. Each ion is treated as a **point charge** surrounded by a continuous dielectric medium (the solvent) of dielectric constant D .
3. Every ion is surrounded by an **ionic cloud** (ionic atmosphere) of net opposite charge, which stabilises the ion electrostatically.
4. The distribution of ions in the atmosphere obeys the **Boltzmann distribution**: $n_i = n_i^0 \exp(-z_i e \psi / k_B T)$, where ψ is the local electrostatic potential.
5. The potential ψ satisfies the **Poisson–Boltzmann equation**. Linearisation (valid for $|z_i e \psi| \ll k_B T$, i.e., low concentrations) gives the **linearised Poisson–Boltzmann** (Debye–Hückel) equation.
6. Ion–ion interactions beyond the central reference ion are described statistically; short-range (excluded volume) effects are neglected.

Debye Length (Thickness of Ionic Cloud) κ^{-1}

The **Debye length** κ^{-1} characterises the effective radius of the ionic atmosphere:

$$\kappa^{-1} = \left(\frac{\varepsilon_0 D k_B T}{2 N_A e^2 I} \right)^{1/2}$$

In SI units with I in mol m^{-3} ; for water at 25°C ($D = 78.5$):

$$\kappa^{-1} \approx \frac{0.304}{\sqrt{I(\text{mol L}^{-1})}} \text{ nm}$$

Dependence of κ^{-1} on parameters:

- $\kappa^{-1} \propto I^{-1/2}$: **increases** at lower ionic strength (dilute solution $\rightarrow$ more diffuse cloud).
- $\kappa^{-1} \propto T^{1/2}$: **increases** with temperature (thermal motion disperses the cloud).
- $\kappa^{-1} \propto D^{1/2}$: larger in high-dielectric solvents (water vs. organic solvents).

- $\kappa^{-1} \propto z^{-1}$: **decreases** for higher-valence electrolytes (stronger electrostatic compression of the cloud).

Worked Example: Debye Length

A uni-univalent electrolyte at 0.10 M in water at 25°C ($D = 78.5$) has $\kappa^{-1} = 0.964$ nm. Find κ^{-1} at 0.001 M.

Since $\kappa^{-1} \propto I^{-1/2} \propto c^{-1/2}$:

$$\frac{\kappa^{-1}(0.001)}{\kappa^{-1}(0.10)} = \sqrt{\frac{0.10}{0.001}} = \sqrt{100} = 10$$

$$\kappa^{-1}(0.001) = 0.964 \times 10 = \boxed{9.64 \text{ nm}}$$

Dimensions of κ^{-1} : From the formula, $[\kappa^{-1}]^2 = [\varepsilon_0 D R T] / [e^2 N_A I]$. Using SI: $[\varepsilon_0] = \text{C}^2 \text{N}^{-1} \text{m}^{-1}$, $[R T] = \text{J mol}^{-1} = \text{N m mol}^{-1}$, $[e^2 N_A I] = \text{C}^2 \text{mol}^{-1} \cdot \text{mol m}^{-3} = \text{C}^2 \text{m}^{-3}$. Therefore:

$$[\kappa^{-1}]^2 = \frac{\text{C}^2 \text{N}^{-1} \text{m}^{-1} \cdot \text{N m}}{\text{C}^2 \text{m}^{-3}} = \frac{\text{m}^0}{\text{m}^{-3}} = \text{m}^2 \quad \Rightarrow \quad [\kappa^{-1}] = \text{m (length)} \quad \checkmark$$

Debye–Hückel Limiting Law (DHLL)

Derivation

Starting from the linearised Poisson–Boltzmann equation, the electrostatic potential around a reference ion of charge $z_i e$ at distance r is:

$$\psi(r) = \frac{z_i e}{4\pi\varepsilon_0 D} \cdot \frac{e^{-\kappa r}}{r}$$

The electrostatic contribution to the chemical potential of the ion (relative to ideal behaviour) is obtained by a charging process (Guntelberg charging):

$$\mu_i^{\text{el}} = \frac{N_A z_i^2 e^2}{8\pi\varepsilon_0 D} \cdot (-\kappa) \quad (\text{leading term, for point charges})$$

Using $\mu_i = \mu_i^\circ + RT \ln a_i = \mu_i^\circ + RT \ln m_i + RT \ln \gamma_i$ and the Debye–Hückel result:

$$RT \ln \gamma_i = -\frac{N_A z_i^2 e^2 \kappa}{8\pi\varepsilon_0 D k_B T} \cdot k_B T = -\frac{N_A z_i^2 e^2}{8\pi\varepsilon_0 D} \cdot \kappa$$

Since $\kappa \propto I^{1/2}$, and collecting all constants into A (which depends on T , D , and solvent):

$$\log_{10} \gamma_i = -A z_i^2 \sqrt{I}$$

For the mean activity coefficient of electrolyte $\text{M}_\nu^{z+} \text{X}_\nu^{z-}$:

$$\boxed{\log_{10} \gamma_{\pm} = -A |z_+ z_-| \sqrt{I}}$$

where for water at 25°C: $A = 0.509 (\text{mol L}^{-1})^{-1/2}$ (or $A = 0.509 \text{ kg}^{1/2} \text{mol}^{-1/2}$ in molality scale). This is the **Debye–Hückel Limiting Law (DHLL)**.

Why It Is Called the “Limiting Law”

The DHLL is strictly valid only in the **limit of infinite dilution** ($I \rightarrow 0$) because:

1. The linearisation of the Boltzmann factor requires $|z_i e \psi| \ll k_B T$, which fails at higher concentrations where the ionic potential is stronger.
2. Ions are treated as **point charges** (no finite size/excluded volume).
3. **Ion–ion correlations** beyond the mean-field approximation are neglected.
4. **Dielectric saturation** (reduction of D near highly charged ions) is ignored.
5. **Ion association** (formation of ion pairs) at higher concentrations is not considered.

Applicability and Failure of the DHLL

	Applicability	Failure
Concentration	$I < 0.01 \text{ mol L}^{-1}$	$I > 0.01 \text{ mol L}^{-1}$
Electrolyte type	1:1 electrolytes best	2:2, 3:1 fail sooner
Dielectric constant	High- D solvents (water)	Low- D solvents
Ion size	Point charge model	Real ions have finite size

Extensions of the DHLL

Debye–Hückel Extended Law (DHEL)

To account for **finite ion size** a (closest approach distance):

$$\log_{10} \gamma_{\pm} = \frac{-A|z_+ z_-| \sqrt{I}}{1 + Ba\sqrt{I}}$$

where $B = \kappa/\sqrt{I}$ is a constant ($B \approx 3.28 \times 10^9 \text{ m}^{-1}$ in water at 25°C) and a is the mean ionic diameter. This equation fits data well up to $I \approx 0.1 \text{ mol L}^{-1}$.

Semi-Empirical Extension (Davies Equation)

For even higher concentrations, a semi-empirical linear term bI is added:

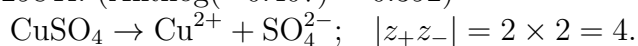
$$\log_{10} \gamma_{\pm} = \frac{-A|z_+ z_-| \sqrt{I}}{1 + \sqrt{I}} + bI$$

where $b \approx 0.30$ for water at 25°C (Davies equation, valid up to $I \approx 0.5 \text{ mol L}^{-1}$). The $+bI$ term accounts for:

- Short-range ion–solvent interactions (salting-out/salting-in effects).
- Hydration of ions which effectively reduces the free water available.
- Ion-pair formation and higher-order correlations.

Worked Example: Mean Activity Coefficient

Problem: Calculate the mean activity coefficient of a 0.010 m aqueous CuSO_4 solution at 298 K. (Antilog(-0.407) = 0.392)



$$I = \frac{1}{2}[0.010(4) + 0.010(4)] = 0.040 \text{ mol kg}^{-1}$$

$$\log_{10} \gamma_{\pm} = -0.509 \times 4 \times \sqrt{0.040} = -0.509 \times 4 \times 0.2000 = -0.407$$

$$\gamma_{\pm} = \text{antilog}(-0.407) = \boxed{0.392}$$

Consequences of Ion–Ion Interaction

Change in Properties of the Reference Ion

In an electrolyte solution, the ionic atmosphere around a reference ion changes its properties in two important ways:

1. **Change in chemical potential (electrostatic energy):** The oppositely charged ionic cloud lowers the electrostatic potential energy of the reference ion. The ion's chemical potential decreases:

$$\mu_i = \mu_i^{\circ} + RT \ln m_i + RT \ln \gamma_i, \quad \gamma_i < 1$$

The term $RT \ln \gamma_i < 0$ represents the stabilisation energy due to the ionic atmosphere. Equivalently, the activity $a_i = \gamma_i m_i < m_i$ (the ion “behaves as” a lower concentration than it actually has).

2. **Change in mobility (retardation of transport):** The ionic atmosphere exerts two retarding forces on a moving reference ion:
 - **Asymmetry (Relaxation) effect:** When the reference ion moves, its ionic cloud cannot instantly rearrange. The cloud lags behind, creating an asymmetric charge distribution. The denser part of the cloud behind the ion exerts a backward electrostatic force, retarding its motion.
 - **Electrophoretic effect:** The ionic cloud moves in the *opposite* direction to the reference ion under the applied field. This sets up a counter-flow of solvent (electrophoretic flow), which the reference ion must move through — increasing the effective viscous drag.

Transport in Electrolyte Solutions

Three Transport Processes

Process	Driving Force	Measure
Diffusion	Concentration gradient (∇c)	Diffusion coefficient D_{diff}
Electric conduction	Electric potential gradient ($\nabla \phi$)	Molar conductivity Λ_m
Viscosity	Velocity gradient (∇v)	Viscosity coefficient η

Molar Conductivity and Its Concentration Dependence

The **molar conductivity** $\Lambda_m = \kappa_{\text{cond}}/c$ (where κ_{cond} is the conductivity) decreases with increasing concentration because:

1. The relaxation (asymmetry) effect increases with c .
2. The electrophoretic effect increases with c .
3. Both are proportional to $\kappa \propto \sqrt{I} \propto \sqrt{c}$.

The **Kohlrausch empirical law**:

$$\Lambda_m = \Lambda_m^0 - K\sqrt{c}$$

where Λ_m^0 is the **limiting molar conductivity** (at infinite dilution) and K is a constant. Plotting Λ_m vs. $\sqrt{c}$ gives a straight line; extrapolation to $\sqrt{c} \rightarrow 0$ gives Λ_m^0 .

Kohlrausch's Law of Independent Migration of Ions

At infinite dilution, each ion migrates *independently*:

$$\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$$

where $\lambda_{\pm}^0$ are the **limiting ionic conductivities**.

Debye–Hückel–Onsager (D-H-O) Equation

Physical Basis

Onsager gave a rigorous derivation of the coefficient K in Kohlrausch's law by calculating the *quantitative* contributions of the relaxation and electrophoretic effects.

Electrophoretic contribution: When the central ion moves in direction $+x$ under field $\mathbf{E}$, the ionic cloud moves in direction $-x$. The cloud carries solvent with it, so the effective medium for the central ion is moving. The electrophoretic retardation to the drift velocity:

$$\Delta v_{\text{EP}} = -\frac{z_i e \kappa}{6\pi\eta_0} \cdot E$$

Contribution to Λ_m :

$$\Delta\Lambda_{\text{EP}} = -\frac{e\kappa}{6\pi\eta_0} = -B_1\sqrt{c}$$

Relaxation (asymmetry) contribution: The asymmetric ionic cloud exerts a backward force. The relaxation time τ for the cloud to reform is finite; during τ , the ion has moved forward, leaving a cloud concentrated behind it. Onsager's result:

$$\Delta\Lambda_{\text{rel}} = -\frac{q}{1+\sqrt{q}} \cdot \frac{e^2\kappa}{12\pi\epsilon_0 D k_B T} \cdot \Lambda_m^0 = -B_2\Lambda_m^0\sqrt{c}$$

where q depends on the transference numbers.

The D-H-O Equation

Combining both contributions:

$$\Lambda_m = \Lambda_m^0 - (B_1 + B_2\Lambda_m^0)\sqrt{c}$$

$$\boxed{\Lambda_m = \Lambda_m^0 - (A + B\Lambda_m^0)\sqrt{c}}$$

where, for water at 25°C: $A = 60.20 \text{ S cm}^2\text{mol}^{-1}(\text{mol L}^{-1})^{-1/2}$ and $B = 0.2289 (\text{mol L}^{-1})^{-1/2}$

(In some texts written as $\Lambda_m = \Lambda_m^0 - (A + B\Lambda_m^0)\sqrt{c}$ with $A = 0.228$ and $B = 60.19$ — conventions differ; the form matches Kohlrausch's $\Lambda_m = \Lambda_m^0 - K\sqrt{c}$.)

Significance of the D-H-O Equation

1. Provides a **theoretical derivation** of Kohlrausch's empirical $\sqrt{c}$ law.
2. The coefficient $K = A + B\Lambda_m^0$ depends on solvent properties and Λ_m^0 only — confirming that the slope of Λ_m vs. $\sqrt{c}$ is determined by solvent and limiting conductivity alone.
3. Works well for **strong electrolytes** at low concentration ($c < 0.01 \text{ mol L}^{-1}$ for 1:1 electrolytes).
4. Fails for **weak electrolytes** (degree of dissociation varies with concentration; must use Ostwald dilution law instead) and at higher concentrations where higher-order effects matter.

Role of Degree of Dissociation and Ion Association

For the D-H-O equation to hold, the electrolyte must be **fully dissociated**. If there is **ion association** (formation of ion pairs, e.g., in low- D solvents), the *effective* concentration of free ions is αc (where α is degree of dissociation), and the D-H-O equation requires modification:

$$\Lambda_m = \alpha\Lambda_m^0 - (A + B\alpha\Lambda_m^0)\sqrt{\alpha c}$$

A plot of Λ_m vs. $\sqrt{c}$ will then show **upward curvature** (concave up), indicating ion association. Only when $\alpha = 1$ (complete dissociation) is the D-H-O plot linear.

Worked Example: D-H-O Calculation

Problem: At 25°C, $A = 0.228$, $B = 60.19 \text{ S cm}^2 \text{ mol}^{-1}$. Calculate Λ_m for 0.01 M HCl. Given: $\lambda^0(\text{H}^+) = 349.8 \text{ S cm}^2 \text{ mol}^{-1}$, $\lambda^0(\text{Cl}^-) = 76.35 \text{ S cm}^2 \text{ mol}^{-1}$.

Step 1:

$$\Lambda_m^0 = 349.8 + 76.35 = 426.15 \text{ S cm}^2 \text{ mol}^{-1}$$

Step 2:

$$K = A + B\Lambda_m^0 = 0.228 + 60.19 \times \frac{426.15}{1000} \approx 0.228 + 25.66 = 25.89$$

(Here B is in units where Λ_m^0 in $\text{S cm}^2 \text{ mol}^{-1}$ and c in mol L^{-1} ; adjust as per the convention used.)

Step 3: Using $\Lambda_m = \Lambda_m^0 - (A + B\Lambda_m^0)\sqrt{c}$:

$$\begin{aligned} \Lambda_m &= 426.15 - (0.228 + 0.2289 \times 426.15)\sqrt{0.01} \\ &= 426.15 - (0.228 + 97.54) \times 0.1 = 426.15 - 97.77 \times 0.1 = 426.15 - 9.78 = \boxed{416.37 \text{ S cm}^2 \text{ mol}^{-1}} \end{aligned}$$

Wien Effect and Debye–Falkenhagen Effect**Wien Effect (High Electric Field)**

At very high applied electric fields ($E > 10^6 \text{ V m}^{-1}$), the conductivity of an electrolyte solution **increases above its low-field value** and approaches Λ_m^0 .

Explanation:

- The high field drives the ion so fast that its **ionic cloud has no time to form or reform** around it.
- The relaxation time τ of the ionic atmosphere satisfies $\tau \sim 10^{-8} \text{ s}$; at high field, the time the ion spends near any given point $\ll \tau$, so the cloud is essentially “left behind”.
- With no ionic cloud: *no* relaxation effect, *no* electrophoretic retardation.
- Hence $\Lambda_m \rightarrow \Lambda_m^0$ at very high field.

Concentration and valence dependence: The Wien effect is *more pronounced* for:

- Higher concentration (the cloud is denser, and its removal causes a bigger increase in Λ_m).
- Higher valence electrolytes (stronger ionic atmosphere, bigger effect when it is destroyed).

Debye–Falkenhagen Effect (High-Frequency AC)

At very high frequency AC fields ($\nu > 10^7 \text{ Hz}$), the conductivity also **increases toward** Λ_m^0 .

Explanation:

- At high frequency, the field reverses direction faster than the ionic cloud can relax (i.e., the oscillation period $\ll \tau_{\text{relax}}$).
- The ionic cloud remains **spherically symmetric** (centred on the ion) because it cannot follow the rapid oscillation of the central ion's position.
- A symmetric cloud exerts *no net* relaxation force on the central ion.
- The electrophoretic effect is also diminished (counter-flow of the cloud averages out over a cycle).
- Result: Λ_m increases and approaches Λ_m^0 .

Note: Both the Wien effect and the Debye–Falkenhagen effect serve as *experimental evidence* validating the ionic atmosphere model of the Debye–Hückel theory.

Non-Ideal Behaviour and Chemical Potential of an Ion

Change in Chemical Potential due to Ion–Ion Interaction

The electrostatic energy per mole of reference ions of type i in the ionic atmosphere:

$$\mu_i^{\text{el}} = N_A \cdot \frac{z_i e \cdot (-z_i e \kappa)}{8\pi\epsilon_0 D} = -\frac{N_A z_i^2 e^2 \kappa}{8\pi\epsilon_0 D}$$

This is always **negative** (stabilisation): the ionic atmosphere has a charge opposite to the central ion, reducing its potential energy.

Connecting to the activity coefficient:

$$RT \ln \gamma_i = \mu_i^{\text{el}} = -\frac{N_A z_i^2 e^2 \kappa}{8\pi\epsilon_0 D} < 0 \quad \implies \quad \gamma_i < 1$$

Mean Activity Coefficient of CuSO_4 (Calculation)

From above (Section 6 worked example):

$$\gamma_{\pm}(\text{CuSO}_4, 0.010 \text{ m}) = \boxed{0.392}$$

Electrophoretic and Relaxation Effects: Quantitative

Electrophoretic Effect

Consider a positive reference ion moving in field $\mathbf{E}$ (in $+x$ direction). Its ionic cloud (net negative) is dragged in the $-x$ direction. The cloud carries solvent molecules with it (electro-osmosis), creating a *counter-flow* of solvent. The reference ion effectively moves against a headwind.

Retardation velocity:

$$\Delta v_{\text{EP}} = -\frac{z_+ e \kappa}{6\pi\eta} E$$

Contribution to molar conductivity:

$$\Delta \lambda_+^{\text{EP}} = -\frac{z_+ e^2 \kappa}{6\pi\eta} \cdot \frac{N_A}{1000} \propto -\sqrt{c}$$

Relaxation (Asymmetry) Effect

When the ion moves, its cloud is distorted: it is denser *behind* the ion (not yet dispersed) and less dense *ahead* (not yet formed). This asymmetric cloud exerts a net backward electrostatic force on the moving ion, slowing it.

Net drift velocity of an ion subject to both effects:

$$v_{\text{drift}} = \frac{z_i e E}{f_0} \left(1 - \frac{q}{1 + \sqrt{q}} \cdot \frac{e^2 \kappa}{12 \pi \varepsilon_0 D k_B T} \right) - \frac{z_i e^2 \kappa E}{6 \pi \eta f_0}$$

where $f_0 = 6 \pi \eta r_i$ (Stokes friction coefficient) and q depends on transference numbers. The first correction is the relaxation effect; the second is the electrophoretic effect.

Experimental Evidence for the Ionic Atmosphere Model

Two experimental observations validate the Debye–Hückel ionic atmosphere model:

1. **Wien effect:** Conductance increases toward Λ_m^0 at high applied field (ion outruns its cloud) — direct confirmation that the ionic atmosphere retards the ion under normal conditions.
2. **Debye–Falkenhagen effect:** Conductance approaches Λ_m^0 at very high AC frequency — confirms that the ionic cloud has a finite relaxation time and cannot follow rapid field oscillations.

Both observations are quantitatively consistent with the Debye–Hückel theory and the predicted relaxation time of the ionic atmosphere.

Key Formulae Summary

Quantity	Expression
Ionic strength	$I = \frac{1}{2} \sum_i c_i z_i^2$
Mean activity coefficient (DHLL)	$\log_{10} \gamma_{\pm} = -A z_+ z_- \sqrt{I}$
DHLL constant (water, 25°C)	$A = 0.509$
Extended DHL	$\log_{10} \gamma_{\pm} = -A z_+ z_- \sqrt{I} / (1 + Ba\sqrt{I})$
Debye length	$\kappa^{-1} = (\epsilon_0 D k_B T / 2 N_A e^2 I)^{1/2}$
Debye length (water, 25°C)	$\kappa^{-1} \approx 0.304 / \sqrt{I} \text{ nm}$
Kohlrausch empirical law	$\Lambda_m = \Lambda_m^0 - K\sqrt{c}$
Kohlrausch law (independent ions)	$\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$
D-H-O equation	$\Lambda_m = \Lambda_m^0 - (A + B\Lambda_m^0)\sqrt{c}$
Activity coefficient correction	$RT \ln \gamma_i = -N_A z_i^2 e^2 \kappa / (8\pi \epsilon_0 D)$

Most Frequently Asked PYQ Topics (Unit 5):

1. Why are electrolyte solutions non-ideal? Consequences of ion–ion interaction.
2. Define ionic strength; calculate I for mixed electrolyte solutions.
3. State the DHLL; derive the expression for $\log \gamma_{\pm}$; discuss applicability and failure (at least 5 reasons for failure).
4. What modifications have been made to the DHLL for concentrated solutions?
5. Derive the Debye–Hückel–Onsager equation; discuss relaxation and electrophoretic effects.

6. What is the Debye length κ^{-1} ? Prove it has dimensions of length. Show how it varies with concentration, temperature, valence.
7. Two experimental proofs of the ionic atmosphere model (Wien effect and Debye–Falkenhagen effect).
8. Explain why conductance increases at high field and high frequency AC.
9. Describe the electrophoretic and relaxation effects on ionic mobility.
10. Calculate the mean activity coefficient of an electrolyte using DHLL.
11. Why is DHLL more reliable for 1:1 than 2:2 electrolytes?

Unifying Picture of Unit 5: Every result in this unit traces back to one physical fact: **each ion is surrounded by an ionic cloud of opposite charge** (the Debye–Hückel model). This single picture explains:

- $\gamma_{\pm} < 1$ (cloud stabilises ion $\rightarrow$ lower chemical potential).
- $\Lambda_m < \Lambda_m^0$ (cloud retards motion via relaxation and electrophoretic effects).
- Wien effect (high field destroys cloud $\rightarrow \Lambda_m \rightarrow \Lambda_m^0$).
- Debye–Falkenhagen effect (high frequency prevents cloud asymmetry $\rightarrow \Lambda_m \rightarrow \Lambda_m^0$).
- DHLL: $\log \gamma_{\pm} \propto -\sqrt{I}$ (cloud energy $\propto \kappa \propto \sqrt{I}$).
- D-H-O: $\Lambda_m = \Lambda_m^0 - K\sqrt{c}$ (retardation $\propto \kappa \propto \sqrt{c}$).